

LUCAS BARBOSA

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MISSION & PRINCIPLES

The entire economy was built on *pre-AI* assumptions. I want to rebuild it. My focus is AI-native operations: going into organisations and reconstructing them from the ground up on infrastructure that treats intelligence as a first-class primitive, not a bolt-on. I am a technologist and an operator in equal measure, and I communicate across both worlds because the gap between them is where most transformations die. I work with extreme intensity on problems that matter, and I don't count hours when the work counts. Writing: lucasbarbosa.net/posts ↗.

PROFESSIONAL EXPERIENCE

oNex AI – Co-Founder, CTO, CPO

2024 – 2026

Full-time – Sydney, Australia

- Engineered a compact proprietary CNN-Transformer for long-range 3D motion classification with **on-device inference at MFLOP scale** across smartphones and watches achieving **95% macro F_1** .
- Developed the end-to-end IMU data platform and collection engine for multi-sensor 3D streams, with robust schema, labeling, and experiment tracking; Led data collection efforts operationally.
- Implemented the backend gRPC mesh connecting Python inference and TypeScript APIs over HTTP/2, achieving intra-region **p50 7 ms and p95 22 ms per RPC**.
- Built a Relay-powered React Native app with generated fragments and normalized caching, integrating NitroModules for native C++ signal-processing via a JSI bridge.
- Technical lead on embedding-driven recommendations using all-MiniLM-L6-v2 with ANN retrieval and lightweight re-rankers, delivering **p95 40 ms** from request to 50 ranked items.

PwC AI – Software Engineer, Tech Lead

2022 – 2025

Full-time, part-time – Sydney, Australia

- Technical lead on a flagship Wage Remediation platform delivered in under six weeks, integrating time-series MSSQL and ETL pipelines; **multi-million-dollar incremental cost savings**, CI/CD anomaly detector scanning 100k+ payroll records, and coordinated cross-functional launch.
- Developed SmartAudit a 10k MAU, RabbitMQ-driven audit platform handling 5k msgs/s across 40+ microservices in 12 countries; Cut frontend memory 80% with low-level v8 engine optimisations enabling hundred-page PDFs to render smoothly on memory-constrained browsers.
- Built a fault-tolerant graph-traversal data purge algorithm, operating at terabyte scale, reducing peak Postgres latency 43% and processing load by 67%.
- Promoted to Senior Consultant prior to graduation, reflecting early progression into technical leadership.

Progue – Software Engineer, Tech Lead

2021 – 2022

Part-time – Sydney, Australia

- Built DOCX data-lake and ms latency vector search API **3× accounting workflow speed**.

Cercle – Software Engineer

2021 – 2022

Part-time – Sydney, Australia

- Launched serverless real-time QR-scanning platform (Lambda, API Gateway, DynamoDB) ingesting 500+ scans/day; cut scan errors 40% and drove product roadmap for funding round.

Lattice – Founder, Software Engineer

2015 – 2022

Full-time, part-time – Sydney Australia

- Delivered bespoke software to 5+ clients at age 16 whilst still ranking in the **top 3% academically**.

ACADEMIC EXPERIENCE

UNSW – Honours Researcher supervised by Dr. Will Midgley

2024 – 2025

Full-time – Sydney, Australia

- Thesis title: Real-World Reinforcement Learning for Vision-Language Robotic Manipulation
- Post-trained SOTA VLA policies using online PPO, TD3 and SAC on OOD manipulation tasks, **improving average task success by ~20%**.
- Fine-tuned GR00T-N1.5 with LoRA on a single RTX 5080 and integrated into a 3D-printed tabletop teleoperation manipulator for data collection and evaluation.

PwC AI – Research Lead

2024 – 2025

Full-time, part-time – Sydney, Australia

- Led 6-person team to publish a peer-reviewed trust-verification system for deepfakes; designed game-theoretic incentive model, authored technical whitepaper, and directed prototype UI/UX.
- Co-developed PwC's first industry-academia IP collaboration framework with internal legal counsel.
- Received coverage from [PwC↔](#), [UNSW↔](#), [AccountingTimes↔](#) and other outlets ([1↔](#), [2↔](#), [3↔](#)).

AtomCraft – Undergraduate Researcher supervised by Dr. S. Setayandeh and Dr. P. Burr Jan – Jun 2024

Full-time – Sydney, Australia

- Ran molecular dynamics simulations using LAMMPS on an HPC cluster to evaluate composite shielding candidates for tokamak fusion reactors, under the Tokamak Energy Research Contract.
- Modeled tungsten boride systems and quantified key material properties including coefficient of linear thermal expansion and threshold displacement energy.

EDUCATION

UNSW – B.Eng (Hons) and B.Sc (Computer Science)

2019 – 2025

Full-time – Sydney, Australia

- Startup Link Society, Events Subcommittee Member
- CouchMobile Autonomy Marketing Secretary
- *Coursework*: Data structures & algorithms (99), Computer architecture (87), Applied control systems (89)

Oakhill College – HSC

2013 – 2018

Full-time – Sydney, Australia

- ATAR: 96.80 (top 3%); House Captain
- Co-founded the XYZ Mathematics Club (inaugural student-run academic club)
- National Computer Science School (NCSS) Summer School 2018

PUBLICATIONS

Barbosa, L., Kirshner, S., Kopel, R., Lim, E.T.K. and Pagram, T. (2025), “Toward trustworthy content: the role of challengers, juries and veracity bonds in digital media platforms”, *Industrial Management & Data Systems*, Vol. 125 No. 11, pp. 3024–3048. [\[Paper↔\]](#) [\[Website↔\]](#)

Barbosa, L., Kirshner, S., Kopel, R., Lim, E.T.K. and Pagram, T., (2025). “A New Incentive Model For Content Trust”, arXiv:2507.09972 [\[Paper↔\]](#)

AWARDS & HONOURS

- 99/100 in COMP2521, Data Structures and Algorithms
- 99.1/100 in HMMAN4951, Honours Thesis A
- [UNIHack 2024↔](#) Hackathon Winner
- [Technion Yom Ha'Atzmaut 2025↔](#) Hackathon Winner
- UNSW ProgComp Distinction Prize 2017
- Oakhill College Principal's Excellence Award (top 1% in academics and leadership)

MENTORSHIP

UNSW Couchmobile Autonomy ROS Tutor

2019 – 2020

UNSW Computer Science HSCH1511 Lab Assistant and Mentor

Feb – May 2018

Zero Robotics Maths Tutor

2017 – 2018

Oakhill College XYZ Mathematics Club Founder and Tutor

2017 – 2018

Oakhill College Mutien House House Captain and Mentor

2017 – 2018

TECHNICAL SKILLS

Languages: C/C++, Go, Rust, Python, TypeScript, SQL, Java, Assembly. **AI/ML:** PyTorch, XGBoost, JAX, reinforcement learning, convolutional neural networks, large language models, vision language models, vision language action models. **Optimization and Modeling:** automatic differentiation, gradient-based optimization, probabilistic modeling. **AI Operations:** LLM eval, prompt versioning, workflow orchestration, observability and tracing, cost and latency optimization. **AI Agents:** tool calling, multi-step workflows, planning, memory, guardrails, evaluation, Claude Code, Cursor, OpenClaw. **Web Frameworks:** React, Solid, Astro, Nest, Node, Bun, React Native, Electron. **Cloud:** GCP, Azure, AWS, IaC. **Tooling:** linux, bash, gdb, monorepos, performance profiling.